

Note:

Remember that since this program can be used with all GTEK programmers, you have to take that into consideration when you are looking at the valid commands for this PGMX7 program. When using PINSTALL, remember that the 7128 can only communicate at baud rates up to 1200 and the others at 2400.

Another difference, especially between the model 9000 and the other GTEK programmers is the way that segment records are handled and displayed. On a 7128, 7228 or 7956, you will only see the lower 4 significant nibbles of the "current address" that the programmer thinks it is looking at when programming or reading a file with segment records. Any error messages will have to be massaged to figure the real address the programmer was working on. For example in a file that starts with a segment record of FD00, a *WP ERR @ D000 is really talking about address FD00:0000 (FD000 - F0000 = D000). You can't see the Most Significant Nibble, but the programmer is keeping track of it. A model 9000 will display that 5th nibble, whereas 7128, 7228 and 7956 can't. Versions of the 7128 can't even keep track of addresses above 64K.

```
:02000002FD00FF          ;segment record
:2000000000*WP ERR @ D000 ;FD000-F0000 = D000
i27512>_                  ;returns to command mode
```

YOU CAN ALSO USE THE PGMX SECTION IN YOUR PROGRAMMER MANUAL.
PGMX IS NOT AVAILABLE FOR THE 7128, 7228, 7956.

USING INTERFACE PROGRAM PGMX7

INSTALLATION of PGMX7

PGMX7, sold separately, is a high speed communication program which runs on IBM PC's and AT's. It allows flexible manipulation, transmission and reception of Intel HEX files and binary files.

On the PGMX7 program disk you will have AT LEAST 3 programs: PGMX7.COM, PINSTALL.COM and GHEX.EXE. SEE THE OTHER README FILE ON THE DISK FOR OTHER PROGRAMS. PGMX7 is the program used to communicate with your programmer. PINSTALL is the program that you must run to install the serial drivers in PGMX7 so that you can communicate with the programmer.

If you try to run the PGMX7 program without installing the serial drivers, it will tell you to run the PINSTALL program. Remember that the PGMX7 license is a single user license. If you buy the programmer now and later on wish to purchase the software, be sure to have the CORRECT serial number ready.

Insert GTEK program disk in drive A: and copy the programs to your hard disk with:

```
C>COPY A:*.*
```

This will copy all the programs on the GTEK disk over to the subdirectory that you are logged on to on your hard disk. If you don't have a hard disk, use DISKCOPY or COPY to the B: drive. Refer to the DOS manual for specific instructions on using the COPY command. The desired end result is a backing up of the original GTEK copy. Store the original program disk in a safe place.

Now you should insert the backup copy in the drive A: and/or go to the subdirectory where PINSTALL and PGMX7 are located. You must first run the PINSTALL program to install the serial drivers for PGMX7.

```
C>PINSTALL<cr>
Serial Driver Installation Program Version 9.05
Copyright 1986, 1987 GTEK, INC.
All Rights Reserved, worldwide.
```

```
Enter name of program to be installed -->PGMX7<cr>
```

```
IBM PC/AT or compatible COM1 (IRQ4)
A - 57600 bps (AT only)
B - 28800 bps (AT only)
C - 19200 bps
D - 9600 bps
E - 2400 bps
F - 1200 bps
G - 300 bps
```

```
IBM PC/AT or compatible COM2 (IRQ3)
H - 57600 bps (AT only)
I - 28800 bps (AT only)
J - 19200 bps
K - 9600 bps
L - 2400 bps
M - 1200 bps
N - 300 bps
```

```
Z - no changes
```

```
Enter selection -->L
Select LPT# (1,2,3 or return for lpt1:) -->1
```

```
Do you have a GTEK Super Serial Card on COM2: (Y/N)? - Y
```

```
OK! What port do you want the programmer on (0-7) ? -->2
```

After the copyright and version number appears, you are asked to select a letter which corresponds to the type of installation you wish to perform.

Most people will probably select the "E" selection which would result in PGMX7 being set up to communicate at 2400 BAUD on computer serial port COM1:. The "L" selection would result in the same thing except COM2: would be the serial port selected.

Press return for lpt1:. If you have a GTEK PCSS-8 SS card in your computer you can then install the software to handle selecting the correct port when PGMX7 is run. If you answer N to SS card then you are returned to DOS.

NOTE: YOU SHOULD NOT MAKE ANY SELECTION THAT IS OUT OF THE BAUD RATE RANGE FOR YOUR PROGRAMMER, WHICH IS 1200 FOR THE 7128, AND 2400 FOR THE 7228 AND 7956. YOU CAN'T MAKE THEM COMMUNICATE FASTER THAN THEY WERE INTENDED TO RUN.

IRQ4 is used in conjunction with an interrupt service routine for COM1: when PGMX7 is invoked if you installed it for COM1:. This is a hardware line on your PC to give the system an interrupt when ever a character is received. If you know that something else in your com-puter is using this hardware interrupt line, then you should use the other com line, which uses IRQ3 (COM2:).

IRQ3 is also used in the same manner for COM2: when PGMX7 is invoked if you installed it for COM2:. If you know something in your system uses IRQ3 for interrupts, then you must use the other com port.

See the example for C>PGMX7<cr> later in the manual.

OPERATION

PGMX7 is a "command driven program" as opposed to a "MENU driven program" which means that everything you do is done by entering a "command" on the command line instead of "selecting" the command from a menu. This makes the program very fast when you have learned what the commands are.

In most cases the commands are exactly the same command as what the programmer is expecting, so the selection of the command is somewhat intuitive.

There are a 2 ways that commands may be given to PGMX7:

- 1- From the PC or MS DOS command line.
- 2- From within PGMX7.

Commands executed from DOS return to DOS upon completion. Commands executed from within PGMX7 return to PGMX7 upon completion. Command lines may be entered from within PGMX7 by depressing control F.

EXAMPLES:

C>PGMX7<cr>

Enter PGMX7 and establish communication with the programmer (assuming everything is hooked up properly).

C>PGMX7 FILENAME<cr>

Results in communication being established with the programmer and sending FILENAME.HEX (Intel Hex Format) from the disk to the programmer. When PGMX7 is through, you are returned to the DOS

system prompt.

C>PGMX7 FILENAME [OPTIONS]<cr>

Results in PGMX7 establishing communication with the programmer, and then performing according to selected options.

Programming the eprom in binary or Intel Hex format or Reading the eprom in the same formats may be accomplished by giving the proper options. OPTIONS are always enclosed in square brackets and separated by comma's. Invalid commands result in an appropriate and descriptive ERROR message.

VALID OPTIONS

R - read file. (default is program mode)
%xxxx - binary mode select (default is HEX)
@xxxx-xxxx eprom bounds
Mx - menu selection
Tx - toggle command (3 max on command line)
V - verify erasure
U - blank check

MORE EXAMPLES

PGMX7<cr> from the DOS command line establishes communication with the programmer, and after log-on displays the Programmer Command Prompter, which is the currently selected eprom type.

LOG ON MESSAGE EXAMPLE

Medium Speed Eprom Programmer Communications Package Version 9.05
Copyright 1983, 1984, 1986, 1987 GTEK, INC.
All Rights Reserved, worldwide.
I/O Hardware Driver Vers 1.04 - IBM PC/XT/AT
Serial port - COM22:, 2400 bps
Printer port - LPT1:

GTEK, INC.
MODEL 7228 V7.11
COPYRIGHT 1983, 1986
xxxx>_

The programmer is ready and waiting for a command at this point.

<iI27128A>TN<cr>
C000
<iI27128A>_

Results in the programmer giving you a 16 bit addition of all the 8 bit bytes of the part without carry, that is in the socket. A blank 27128 gives you C000 for the checksum.
<iI27128A>(control)

Control- generally means to press and hold the CONTROL key on your keyboard and press a command letter. Valid command letters are P, F and C. The ESCape key is also a valid control command key, but you do not hold the control key down to press ESC. The ESC key is a valid control character already. The escape control

command may also be obtained by pressing CONTROL [on the IBM keyboard or by holding down the ALT key and entering 27 on the numeric keypad. Pressing and holding the CONTROL key is represented by a caret and the letter that must also be pressed, eg. ^C.

The definitions of the CONTROL commands are:

^P -start sending / stop sending (toggle) data simultaneously to the printer.

^F -enter a command line. Examples follow.

^C -Abort most programmer commands and return to the DOS command prompt or PGMX7. This command will work even though you may be in the process of programming, reading, verifying, etc., an eprom.

ESC or ^[- Escape from program. This command is used as an alternative to control-alt-del and is not normally used. This is an EMERGENCY command and the results could be unpredictable.

USING control F

<2716>^F

Enter Command line >FILENAME [@0-1FF,V,TN

Results in PGMX7 doing a blank check on the eprom between 0 and 1FF inclusive. Then FILENAME.HEX is opened and any hex data falling between the specified boundaries is sent. During data transfer, PGMX7 displays the load addresses of the hex records that it is sending. Finally, the checksum is calculated between the specified addresses and displayed.

The options are always set off by an opening square bracket ([) and the ending square bracket (]) is optional. Invalid commands result in an error message and a return to the programmer command prompt.

DEFINITIONS

Please note that the listed commands are generally passed on to the programmer unchanged except for the order in which they appear in the command line. PGMX7 will send the commands specified to the programmer in the following order:

- 1 - menu command
- 2 - toggle commands (except TN is done last)
- 3 - blank check or verify erasure
- 4 - program or read
- 5 - checksum (tn)

Some commands, particularly the "R" command, work differently from the programmer command "R". The "%" and the "@" command are not valid commands for the programmer except on the command line. They are used to give PGMX7 information, not the programmer. You may not specify any command more than once inside the brackets except the toggle commands, and you are allowed a maximum of 3 of those.

ssss = starting address, ascii-hex characters
eeee = ending address, ascii-hex characters
oooo = offset amount, ascii-hex characters

Delimiter = a dash (-), a comma (,), a space (), a carriage return (ascii character number 0dh), or a line feed (ascii character number 0ah). Carriage return and line feed are represented by a <cr> or <lf>.

FILENAME = a valid DOS filename to be used by PGMX7 to look for a file on the disk. In case that a percent (%) sign was specified then the filename specified will be taken literally. In other words you must be explicit and give the extension of the filename also. If the percent sign was not specified then PGMX7 will automatically supply a .HEX extension and look for a .HEX even if you specified an extension.

EXT = a valid DOS extension for the filename in your directory. You are allowed to use any extension you wish here, (in the binary % mode) and the data will be sent to the programmer UNCHANGED. The EXT will only be valid when you have specified a percent sign (%) within the brackets.

AND REMEMBER!

The effective addressing range of a device is determined by it's size. If a 2K byte device is being used, then it only has 11 significant address lines and only the lowest 11 bits of the address field are significant. Thus, as far as the 9000 is concerned, 000H is equivalent to 800H or F000H in a 2K device.

VALID COMMANDS FOR PGMX7

- 1- Any valid programmer command except OI, OM, OT, R
- 2- @ssss-eeee<cr>

An @ symbol followed by the starting address (ssss) followed by a dash (-) followed by the ending address (eeee) will cause the incoming data from the Binary or Intel Hex File to be sent to the specified locations. In the case of a binary file (specified by a % on the same command line only), the @ symbol means that the data specified by the % sign will go to the ssss-eeee specified by the @ sign within the eprom. In the case of an Intel Hex file (no %), the @ symbol means that PGMX7 will search the Intel Hex file for data located between the start address (ssss) and the end address (eeee) and send that data to the same locations within the eprom.

- 3- %oooo

A percent sign (%) followed by an offset (you may omit specifying an offset of 0, but PGMX7 may warn you that you did not specify it, just in case you forgot) will cause PGMX7 to treat the EXTension you specified literally (and not add a .HEX extension).

EXAMPLES:

To program 3 2716's from a binary file that contains 1093H

bytes:

```
<xxxx>MB
<2716>^F
Enter Command line -->TEST.BIN[%0,@0-7FF<cr>
```

Causes PGMX7 to look for a file called TEST.BIN on the disk, and when found start sending from relative offset location 0 within TEST.BIN to locations 0 through 7FFh within the eprom. The number of bytes sent is the number of bytes between 0 to 7FFh inclusive. If you don't specify boundaries, you will "Wrap Around" to location 000H at location 800H because you are still sending data to the programmer through PGMX7.

```
<2716>^F
Enter command line -->TEST.BIN[%800,@0-7FF<cr>
```

Causes PGMX7 to look for a file called TEST.BIN on the disk, and when found start sending from relative offset 800H from within TEST.BIN to locations 0 to 7FFh within the eprom.

```
<2716>^F
Enter command line ->TEST.BIN[%1000,@0-7FF<cr>
```

Causes PGMX7 to look for a file called TEST.BIN on the disk and when found start sending from relative offset 1000H from within the TEST.BIN to locations 0 through 7FFh within the eprom. However, the program will terminate when it encounters the end of the file you are sending from since there are only 94H bytes left in the file TEST.BIN left to send.

Automating the process could be accomplished with a batch file such as this:

```
TEST.BAT
ECHO OFF
PGMX7 test.bin[mb,V,%0,@0-7ff,tn
ECHO remove eprom, insert new blank
PAUSE
PGMX7 test.bin[V,%800,@0-7ff,tn
ECHO remove eprom, insert new blank
PAUSE
PGMX7 test.bin[V,%1000,@0-7ff,tn
echo now you are done.
```

Reading an eprom to a disk file is accomplished with the 'R' option.

```
PGMX7 filename[r<cr>
```

Results in reading the selected eprom to the Intel hex disk file, FILENAME.HEX.

```
PGMX7 filename[r,%
```

Results in reading the selected eprom to a binary disk file whose name is FILENAME. (no extension was specified.). Note that an offset value included with the % has no meaning during a read

operation.

PGMX7 [tn,ma or ^F[tn,ma from within PGMX7

Results in selecting 2758 (note menu selection has side effect of resetting all toggles) and calculating the checksum.

ADVANCED EXAMPLE

PGMX7 filename[m7,ts,v,tn,@20000-3ffff

Results in selecting 27512, split mode, doing a blank check, programming the eprom with hex data residing between the 20 bit addresses of 20000 and 3ffff inclusive, and calculating it's checksum.

This particular file is big. Don't be afraid that PGMX7 has hung up. It has to check the load addresses of every record in the file, and it would take a minute before it reached records at load address 20000, unless the file was created with an "exotic" compiler in such a manner that segment records with apparently random addresses are placed at apparently random locations every few records in the file. No joke intended.

The boundaries specified cover a 128k range, but the eprom is only 64k. The reason for this is that in the spit mode, the 2 eproms are considered as one eprom of twice the size. However, if an error message is issued during programming in the split mode, the address given by the error message is the physical address in the single eprom.

USAGE OF GHEX.EXE

GHEX.EXE is a program provided for you to be able to convert a binary file into an INTEL.HEX file. This capability is built-in to the PGMX7.COM program, but you may want to use it for convenience.

General usage is:

GHEX filename.ext<cr>

OR

GHEX filename.ext offset

Offset is an ASCII-HEX number that specifies where you want your code to begin in the HEX file.

GHEX filetest.bin

Will result in an INTEL.HEX file being created on your disk by the name filetest.hex. The load addresses begin at 0000H since no offset was specified. GHEX does not destroy the input file.

GHEX filetest.bin AA55

Will result in an INTEL.HEX file being created on your disk by

the name filetest.hex, just like before except the load addresses start at the offset address of AA55H in the file.

USING DEBUG.COM

You may use DEBUG.COM (supplied with PC-DOS) in conjunction with our GHEX.EXE to modify an INTEL.HEX file without worrying about the checksums in the INTEL.HEX file.

The following is a short tutorial to modify a 4K byte INTEL.HEX file with DEBUG. The procedure is to run DEBUG first.

```
DEBUG<cr>
```

```
-_
```

From the - prompter within DEBUG use the N command to specify the name of your INTEL.HEX file.

```
-Nfilename.HEX<cr>
```

```
-_
```

Use the L command to load the hex file with an offset (if it begins at 0000H). You must do this since if it starts loading at 0000H within the segment, it will overwrite your file control block at 5Ch.

```
-L 100<cr>
```

```
-_
```

The CX register now contains the number of bytes read into memory with an offset of 100. You may have to modify the CX register to properly reflect the correct number bytes you must write back to the disk. Remember that this is going to write from CS:CX when you issue the command.

```
-RCX<cr>
```

```
CX: 1000<cr>
```

```
-_
```

Your data is now loaded into the memory of the computer at offset 100H. Use the E command to modify the bytes you need to modify. An example of modifying locations starting at 0A55H with data is shown. Locations A55H through A57H contain FFH.

```
-EA55 01 02 03<cr>
```

```
-_
```

Now specify a new file name to write to the disk with since you can't use an extension of HEX with the file you are writing. You want to call it a BIN or IMG file instead since that is what the data really is anyway.

```
-NNEWFILE.BIN<cr>
```

```
-_
```

Now you can use the Write command to write the new data to the disk. DEBUG will write an exact image of CS:CX bytes to the disk starting at an offset of 0100H bytes.

-W<cr>
Writing 1000H bytes
- _

Now you can use GHEX to make it an INTEL.HEX file or use the
BINARY transfer feature of PGMX7 to write the data to the blank
eprom.

Change.exe

Change.EXE is a program that can convert Motorola S0-S2 files into the
INTEL hex format. Run it for instructions.